بِسۡمِ ٱللَّهِ ٱلرَّحۡمَٰنِ ٱلرَّحِيمِ

(Start the day with the name of Allah)

Welcome To Virtual University

[redacted] AMNA IRSHAD

Time Left 69 sec(s)

CS402:Grand Quiz

Quiz Start Time: 11:31 AM, 23 June 2021

Question # 1 of 30 (Start time: 11:31:11 AM, 23 June 2021) Total Marks: 1

Which one of the following word is not accepted by the regular expression (a+b)*(aa)(a+b)*?

Select the correct option

- aa
- aaaa
- bab
- baaa

Solved By: Waqas Ahmed
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Email: waqasahmeduog@gmail.com

Question # 2 of 30 (Start time: 11:31:44 AM, 23 June 2021)

Total Marks: 1

FA corresponding to an NFA can be built by introducing an empty state for a letter having

Select the correct option

- more than two transitions at certain state
- one transition at certain state
- no transition at certain state

Solved By: Waqas Ahmed
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- two transitions at certain state

CS402:Grand Quiz

Quiz Start Time: 11:31 AM, 23 June 2021

Question # 3 of 30 (Start time: 11:32:19 AM, 23 June 2021) **Total Marks: 1**

Suppose we have FA3 (which is equal to FA1 + FA2), then the final state of FA3 will be declared final if:

Select the correct option

- It corresponds to final states of FA1 only
- It corresponds to final states of both FA1 and FA2
- It corresponds to any of the final states in FA1 or FA2
- It corresponds to final states of FA2 only

Solved By: Waqas Ahmed
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Question # 4 of 30 (Start time: 11:33:01 AM, 23 June 2021)

Total Marks: 1

The recursive method for defining a language has _____ steps.

Select the correct option

- two
- four
- three
- one

Solved By: Waqas Ahmed
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Question # 5 of 30 (Start time: 11:33:52 AM, 23 June 2021) **Total Marks:** 1

Two machines are said to be equivalent if they print the _____ output string when the _____input string is run on them.

Select the correct option

- (●) same, same

Solved By: Waqas Ahmed
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Email: waqasahmeduog@gmail.com

- ( ) different, same
- ( ) unique, different
- ( ) same, different

Question # 6 of 30 (Start time: 11:34:19 AM, 23 June 2021) Total Marks: 1

Kleene's Theorem Part III expresses the relationship between _____.

Select the correct option

- (●) RE and FA

Solved By: Waqas Ahmed
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- ( ) TG and RE
- ( ) FA and RE
- ( ) FA and TG

Question # 7 of 30 (Start time: 11:34:43 AM, 23 June 2021) **Total Marks:** 1

Considering FA1 and FA2 having 2 states each. Now FA1+FA2 can have maximum _____ number of states.

Select the correct option

- None of the given options
- 3
- more than 3

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- 2

CS402:Grand Quiz

Quiz Start Time: 11:31 AM, 23 June 2021

Question # 8 of 30 (Start time: 11:35:05 AM, 23 June 2021)

Total Marks: 1

FA stands for _____.

Select the correct option

- Fixed Automaton
- Functional Automaton
- Finite Automaton

Solved By: Waqas Ahmed
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- False Automaton

Question # 9 of 30 (Start time: 11:35:27 AM, 23 June 2021) **Total Marks:** 1

The language of all strings defined over alphabet set = {x, y} having triple x's or triple y's will have the minimum strings with length of:

Select the correct option

- 2
- 4
- 3
- 1

Solved By: Waqas Ahmed
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Email: waqasahmeduog@gmail.com

Question # 10 of 30 (Start time: 11:35:53 AM, 23 June 2021)

Total Marks: 1

The aa(a+b)*bb is the RE of the language defined over Σ={a,b}.The language must _____.

Select the correct option

- have at least one ba
- have at least one aa and one bb
- have at least one abbb
- have at least two ab

Solved By: Waqas Ahmed
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Question # 11 of 30 (Start time: 11:36:18 AM, 23 June 2021) **Total Marks:** 1

A loop at a state is supposed to be ---------- transition while converting Moore machine into an equivalent Melay machine.

Select the correct option

- (●) Incoming

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- ( ) Outgoing
- ( ) Complex
- ( ) Both incoming and outgoing

Saving

CS402:Grand Quiz

Quiz Start Time: 11:31 AM, 23 June 2021

Question # 12 of 30 (Start time: 11:36:48 AM, 23 June 2021)

Total Marks: 1

In an FA, when there is no path starting from initial state and ending in final state then that FA_____.

Select the correct option

- accept all non empty strings
- does not accept any string

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- accept all strings
- accept null string

CS402:Grand Quiz

Quiz Start Time: 11:31 AM, 23 June 2021

Question # 13 of 30 (Start time: 11:37:16 AM, 23 June 2021)

Total Marks: 1

Let FA3 be an FA corresponding to FA1FA2, then the final state of FA3 must correspond to the final state of

Select the correct option

- FA1 only
- FA1 or FA2
- FA1 and FA2
- FA2 only

Solved By: Waqas Ahmed
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CS402:Grand Quiz

Quiz Start Time: 11:31 AM, 23 June 2021

Question # 14 of 30 (Start time: 11:37:40 AM, 23 June 2021)

Total Marks: 1

Which one of the following word is not accepted by the given regular expression?
(a+b)*bba(a+b)*

Select the correct option

- (●) aaababab

Solved By: Waqas Ahmed
Whatsapp: 03066295623
Email: waqasahmeduog@gmail.com

- ( ) bbabbbbaba
- ( ) babbba
- ( ) abbbbaa

CS402:Grand Quiz

Quiz Start Time: 11:31 AM, 23 June 2021

Question # 15 of 30 (Start time: 11:38:21 AM, 23 June 2021)

Total Marks: 1

Substrings as input letters can be specified on edges in:

Select the correct option

- PDA
- NFA
- TG

Solved By: Waqas Ahmed
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- FA

Question # 16 of 30 (Start time: 11:38:45 AM, 23 June 2021) **Total Marks:** 1

If we have a finite language and the number of states in the FA is n then the maximum number of letters in the each word of the language that will be accepted by the given FA will be:

Select the correct option

- 1
- n+1
- n
- n-1

Solved By: Waqas Ahmed
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Email: waqasahmeduog@gmail.com

Question # 17 of 30 (Start time: 11:39:09 AM, 23 June 2021) **Total Marks:** 1

Suppose we have an alphabet set having four symbols, how many transitions will be there on each state of a finite automaton for any language defined over the given alphabet?

Select the correct option

- ○ 1
- ○ 2
- ○ 3
- ● 4

Solved By: Waqas Ahmed
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Question # 18 of 30 (Start time: 11:39:31 AM, 23 June 2021)

Total Marks: 1

A string will be accepted by an NFA if there exists _____ one successful path.

Select the correct option

- none of the given options
- atmost
- atleast
- maximum

Solved By: Waqas Ahmed
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CS402:Grand Quiz

Quiz Start Time: 11:31 AM, 23 June 2021

Question # 19 of 30 (Start time: 11:40:01 AM, 23 June 2021)

Total Marks: 1

We cannot construct an NFA for the language of _____ defined over alphabet set {a,b}.

Select the correct option

- Palindromes

Solved By: Waqas Ahmed
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- Even even
- odd
- Integers

CS402:Grand Quiz

Quiz Start Time: 11:31 AM, 23 June 2021

Question # 20 of 30 (Start time: 11:40:38 AM, 23 June 2021) **Total Marks:** 1

Which one of the following word is not accepted by the given regular expression?
(0+1)*(00+11)(0+1)*

Select the correct option

- 1010101

Solved By: Waqas Ahmed
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- 0100010
- 0001100
- 1110011

CS402:Grand Quiz

Quiz Start Time: 11:31 AM, 23 June 2021

Question # 21 of 30 (Start time: 11:41:08 AM, 23 June 2021)

Total Marks: 1

Which of the following state is introduced while developing NFA for the closure of an FA?

Select the correct option

- Final state
- An initial state with loop for all letters
- Simply an initial state
- An initial state which should be final as well

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CS402:Grand Quiz

Quiz Start Time: 11:31 AM, 23 June 2021

Question # 22 of 30 (Start time: 11:41:31 AM, 23 June 2021)

Total Marks: 1

There _____ be dead states in NFA.

Select the correct option

- must
- may not

Solved By: Waqas Ahmed
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Email: waqasahmeduog@gmail.com

- should not
- will

Question # 23 of 30 (Start time: 11:42:14 AM, 23 June 2021)

Total Marks: 1

The language having even number of a's and even number of b's defined over S = {a, b} is called _____.

Select the correct option

- (●) EVEN-EVEN

Solved By: Waqas Ahmed
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Email: waqasahmeduog@gmail.com

- ( ) FACTORIAL
- ( ) PALINDROME
- ( ) ODD-ODD

CS402:Grand Quiz

Quiz Start Time: 11:31 AM, 23 June 2021

Question # 24 of 30 (Start time: 11:42:42 AM, 23 June 2021)

Total Marks: 1

Which of the following regular expressions represent same language?

1. (a+ab)*
2. (ba+a)*
3. a*(aa*b)*
4. (a*b*)*

Select the correct option

- 3 and 4
- 1 and 4
- 1 and 3

Solved By: Waqas Ahmed
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- 1 and 2

CS402:Grand Quiz

Quiz Start Time: 11:31 AM, 23 June 2021

Question # 25 of 30 (Start time: 11:43:09 AM, 23 June 2021)

Total Marks: 1

If r1 is a regular expression then (r1)* is _____.

Select the correct option

- a generalized transition graph
- a finite automaton
- also a regular expression

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- a non-deterministic finite automaton

CS402:Grand Quiz

Quiz Start Time: 11:31 AM, 23 June 2021

Question # 26 of 30 (Start time: 11:43:36 AM, 23 June 2021)

Total Marks: 1

The language of all strings defined over alphabet set = {x, y} that ends with same letters will have the maximum length of:

Select the correct option

- (●) infinite

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Email: waqasahmeduog@gmail.com

- ( ) 2
- ( ) 3
- ( ) 1

CS402:Grand Quiz

Quiz Start Time: 11:31 AM, 23 June 2021

Question # 27 of 30 (Start time: 11:44:09 AM, 23 June 2021)

Total Marks: 1

According to theory of automata there are _____ types of languages.

Select the correct option

- One
- Two

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- Four
- Three

CS402:Grand Quiz

Quiz Start Time: 11:31 AM, 23 June 2021

Question # 28 of 30 (Start time: 11:44:36 AM, 23 June 2021)

Total Marks: 1

How many states of a finite automaton will be final for accepting L = {^, b, bb, bbb}?

Select the correct option

- 3
- 2
- 4

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- 1

CS402:Grand Quiz

Question # 29 of 30 (Start time: 11:45:04 AM, 23 June 2021)

FA and _____ are same except that _____ has unique symbol for each transition.

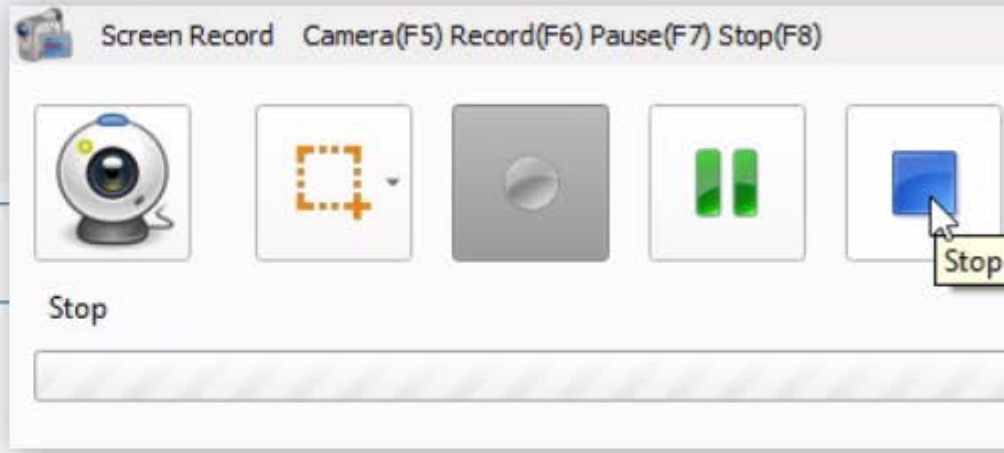
Select the correct option

○	FA,TG
○	NFA,FA
○	NFA,TG
○	GTG,NFA

CS402:Grand Quiz

Question # 30 of 30 (Start time: 11:45:12 AM, 23 June 2021)

If A and B are regular languages, !(A' U B') is:



Select the correct option

- non regular
- regular
- may be regular
- none of the mentioned

[redacted]: ARSHAD ALI

CS402:Grand Quiz

Question # 1 of 30 (Start time: 01:44:40 PM, 23 June 2021)

There _____ a language for which only FA can be built but not the RE.

Select the correct option

- may not be
- cannot be

Solved By: Waqas Ahmed
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- is
- may be

CS402:Grand Quiz

Question # 2 of 30 (Start time: 01:45:50 PM, 23 June 2021)

If we have an NFA having 3 states, and we convert that NFA to an FA. The resultant FA will contains _____ states.

Select the correct option

- ○ 5
- ○ 3
- ○ 2
- ● 4

Solved By: Waqas Ahmed
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CS402:Grand Quiz

Question # 3 of 30 (Start time: 01:46:05 PM, 23 June 2021)

There is no question of accepting any language in:

Select the correct option

- FA
- GTG
- Moore machine
- TG

Solved By: Waqas Ahmed
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CS402:Grand Quiz

Question # 4 of 30 (Start time: 01:46:19 PM, 23 June 2021)

Kleene's Theorem Part II expresses the relationship between _____.

Select the correct option

- RE and FA
- TG and RE
- FA and TG
- FA and RE

Solved By: Waqas Ahmed
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CS402:Grand Quiz

Question # 5 of 30 (Start time: 01:46:36 PM, 23 June 2021)

Which of the following is not a step in elimination of states procedure?

Select the correct option

- Remove states until there is only starting and accepting states
- Unifying all the final states into one using e-transitions
- Get the resulting regular expression by direct calculation
- Unify single transitions to multi transitions that contains union of input

Solved By: Waqas Ahmed
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CS402:Grand Quiz

Question # 6 of 30 (Start time: 01:46:54 PM, 23 June 2021)

Which of the following does not contribute while finding out the length of strings?

Select the correct option

- a+b
- a
- b
- ^

Solved By: Waqas Ahmed
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CS402:Grand Quiz

Question # 7 of 30 (Start time: 01:47:07 PM, 23 June 2021)

n! will be equal to:

Select the correct option

- n*(n-n)!
- n*n
- n*(n-1)
- n*(n-1)!

Solved By: Waqas Ahmed
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CS402:Grand Quiz

Question # 8 of 30 (Start time: 01:47:20 PM, 23 June 2021)

Let S = {aa, bb}, then S* will have the _____ string.

Select the correct option

- aabbbaa
- Λ
- abba
- bbaab

Solved By: Waqas Ahmed
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CS402:Grand Quiz

Question # 9 of 30 (Start time: 01:47:34 PM, 23 June 2021)

Melay machine can have -------- final states.

Select the correct option

- (●) Zero

Solved By: Waqas Ahmed
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- ( ) More than one but finite
- ( ) One
- ( ) More than one but infinite

CS402:Grand Quiz

Question # 10 of 30 (Start time: 01:47:50 PM, 23 June 2021)

The length of output string in case of _____ is one more than the length of corresponding input string.

Select the correct option

- ○ Adding Machine
- ● Moore machine

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- ○ Mealy machine
- ○ Incremental machine

CS402:Grand Quiz

Question # 11 of 30 (Start time: 01:48:04 PM, 23 June 2021)

Which one of the following RE describes the language of strings having exactly double a, defined over
Σ ={a, b}?

Select the correct option

- (a+b)*
- a*+b*
- a*b*
- b*aab*

Solved By: Waqas Ahmed
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CS402:Grand Quiz

Question # 12 of 30 (Start time: 01:48:19 PM, 23 June 2021)

For every three regular expressions R, S, and T, the languages denoted by R(S U T) and (RS) U (RT) are the _____ .

Select the correct option

- (●) Same

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- ( ) None of the given options
- ( ) R(S U T) is Greater
- ( ) Different

CS402:Grand Quiz

Question # 13 of 30 (Start time: 01:48:41 PM, 23 June 2021)

In proving Kleene Theorem II, if three states are connected then middle state is removed by connecting first and third state and writing corresponding RE

Select the correct option

- (●) Concatenation

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- ( ) Asterisk
- ( ) Difference
- ( ) Sum

CS402:Grand Quiz

Question # 14 of 30 (Start time: 01:48:56 PM, 23 June 2021)

According to theory of automata there are _____ types of languages.

Select the correct option

- (●) Two

Solved By: Waqas Ahmed
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- ( ) Four
- ( ) Three
- ( ) One

CS402:Grand Quiz

Question # 15 of 30 (Start time: 01:49:15 PM, 23 June 2021)

Which one of the following word is not accepted by the given regular expression?
aa(a+b)*b

Select the correct option

- aaabab
- abbaab
- aabbabb
- aaaababb

Solved By: Waqas Ahmed
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CS402:Grand Quiz

Question # 18 of 30 (Start time: 01:49:55 PM, 23 June 2021)

All possible combinations of strings of a language including null string is referred as:

Select the correct option

- ○ Multiplication of a language with itself
- ○ Concatenation of a language with itself
- ○ Addition of a language with itself
- ● Kleene star closure of a language

Solved By: Waqas Ahmed
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CS402:Grand Quiz

Question # 19 of 30 (Start time: 01:50:08 PM, 23 June 2021)

Suppose we have an alphabet set having four symbols, how many transitions will be there on each state of a finite automaton for any language defined ov

Select the correct option

○	2
○	1
●	4 Solved By: Waqas Ahmed Whatsapp: 03066295623 Email: waqasahmeduog@gmail.com
○	3

CS402:Grand Quiz

Question # 20 of 30 (Start time: 01:50:21 PM, 23 June 2021)

Which of the following is the bypass and state elimination step in the context of Kleene's theorem part II proof?

Select the correct option

- 2
- 3
- 1
- 4

Solved By: Waqas Ahmed
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CS402:Grand Quiz

Question # 21 of 30 (Start time: 01:50:32 PM, 23 June 2021)

In which of the following machine, the length of output string is the same to that of input string?

Select the correct option

- ○ Finite automaton with output
- ● Mealy machine

Solved By: Waqas Ahmed
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- ○ Non-deterministic finite automaton
- ○ Moore machine

CS402:Grand Quiz

Question # 22 of 30 (Start time: 01:50:46 PM, 23 June 2021)

Every _____ is a _____ as well, but the converse may not be true.

Select the correct option

- TG, GTG
- FA, TG
- FA,GTG
- TG, FA

Solved By: Waqas Ahmed
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CS402:Grand Quiz

Question # 23 of 30 (Start time: 01:51:01 PM, 23 June 2021)

Two FAs are said to be equivalent, if they _____.

Select the correct option

- ○ none of the given options
- ● accept same language

Solved By: Waqas Ahmed
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- ○ accept null string
- ○ accept different language

CS402:Grand Quiz

Question # 24 of 30 (Start time: 01:51:17 PM, 23 June 2021)

Keeping in view the language of all strings ending with 'a', for which symbol we will take a loop on the final state of its transition diagram?

Select the correct option

- ○ c
- ○ b
- ● a

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- ○ d

CS402:Grand Quiz

Question # 25 of 30 (Start time: 01:51:31 PM, 23 June 2021)

Strings x,y,z belongs to Σ^* such that both xz or yz $\in$ L where $L \subseteq \Sigma^*$ are:

Select the correct option

- Both distinguishable and indistinguishable
- Distinguishable
- Indistinguishable
- Undetermined

Solved By: Waqas Ahmed
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CS402:Grand Quiz

Question # 26 of 30 (Start time: 01:51:48 PM, 23 June 2021)

---------- state is not important in Melay machine.

Select the correct option

- Initial — Solved By: Waqas Ahmed
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  Email: waqasahmeduog@gmail.com
- Final
- Combination of
- Empty

CS402:Grand Quiz

Question # 27 of 30 (Start time: 01:52:03 PM, 23 June 2021)

Keeping in view the discussion by Martin, how many states are required to recognize the language of all strings of length 3 or more defined over ∑= {a,b}

Select the correct option

- ○ 14
- ● 15

Solved By: Waqas Ahmed
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- ○ 13
- ○ 16

CS402:Grand Quiz

Question # 28 of 30 (Start time: 01:52:16 PM, 23 June 2021)

Which of the following is an utility of state elimination phenomenon?

Select the correct option

- NFA to DFA
- DFA to Regular Expression

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- DFA to NFA
- All of the mentioned

CS402:Grand Quiz

Question # 29 of 30 (Start time: 01:52:31 PM, 23 June 2021)

Regular languages are closed under the following operations.

Select the correct option

- (●) Union, Concatenation and Closure

Solved By: Waqas Ahmed
Whatsapp: 03066295623
Email: waqasahmeduog@gmail.com

- ( ) Union only
- ( ) Concatenation, Closure only
- ( ) Regular languages are not closed under any operation.

Question # 30 of 30 (Start time: 01:52:48 PM, 23 June 2021)

Which of the following doesn't accept any language?

Select the correct option

- ○ Deterministic finite state machine
- ○ Finite state machine
- ○ Moore machine
- ○ Regular expression

[redacted]ZAIN ASIF

Time Left 72 sec(s)

CS402:Grand Quiz

Quiz Start Time: 12:41 PM, 23 June 2021

Question # 1 of 30 (Start time: 12:41:02 PM, 23 June 2021) Total Marks: 1

Keeping in view the discussion by Martin, how many states are required to recognize the language of all strings of length 3 or more defined over ∑= {a,b}, with 'a' being the third letter from right?

Select the correct option

- 15

Solved By: Waqas Ahmed
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- 14
- 16
- 13

CS402:Grand Quiz

Quiz Start Time: 12:41 PM, 23 June 2021

Question # 2 of 30 (Start time: 12:41:36 PM, 23 June 2021)

Total Marks: 1

If S = {aa, bb} then S* will not contain _____.

Select the correct option

- bbaabbbb
- aaabbb

Solved By: Waqas Ahmed
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- aabbbb
- aabbaa

CS402:Grand Quiz

Quiz Start Time: 12:41 PM, 23 June 2021

Question # 3 of 30 (Start time: 12:41:59 PM, 23 June 2021)

Total Marks: 1

In Moore machine, output is produced over the change of:

Select the correct option

- Transitions and states
- None of the mentioned
- States

Solved By: Waqas Ahmed
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Email: waqasahmeduog@gmail.com

- Transitions

CS402:Grand Quiz

Quiz Start Time: 12:41 PM, 23 June 2021

Question # 4 of 30 (Start time: 12:42:24 PM, 23 June 2021)

Total Marks: 1

How many states of a finite automaton will be final for accepting the only string 'abb'?

Select the correct option

1

Solved By: Waqas Ahmed
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4

3

2

CS402:Grand Quiz

Quiz Start Time: 12:41 PM, 23 June 2021

Question # 5 of 30 (Start time: 12:42:47 PM, 23 June 2021) **Total Marks:** 1

In TG, the string is supposed to be _____ if there is no path for a string from initial to final state.

Select the correct option

- Accepted
- Partially accepted
- Rejected
- Partially rejected

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CS402:Grand Quiz

Quiz Start Time: 12:41 PM, 23 June 2021

Question # 6 of 30 (Start time: 12:43:12 PM, 23 June 2021)

Total Marks: 1

There may be more than one transition for a certain letter on a state in:

Select the correct option

- Moore Machine
- Transition Table
- Non-Deterministic Finite Automata
- Finite Automata

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Question # 7 of 30 (Start time: 12:43:38 PM, 23 June 2021) **Total Marks: 1**

In the context of make NFA for the concatenation of FA1 and FA2 (FA2 accepting null string), which of the following option is correct?

Select the correct option

- (●) Final states in both FAs

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- ( ) Initial states in both FAs
- ( ) FA2 having initial state only
- ( ) FA2 having final state only

CS402:Grand Quiz

Quiz Start Time: 12:41 PM, 23 June 2021

Question # 8 of 30 (Start time: 12:44:02 PM, 23 June 2021)

Total Marks: 1

Consider we have languages L7 and L6. Which of the following represents their concatenation?

Select the correct option

- (●) L7L6
- ( ) L7/L6
- ( ) L7+L6
- ( ) L6*L7

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Question # 9 of 30 (Start time: 12:44:25 PM, 23 June 2021)

Total Marks: 1

A transition graph is converted into a(n) _____ in order to obtain regular expression.

Select the correct option

- NFA-^
- FA
- NFA
- GTG

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CS402:Grand Quiz

Quiz Start Time: 12:41 PM, 23 June 2021

Question # 10 of 30 (Start time: 12:44:52 PM, 23 June 2021)

Total Marks: 1

There is no compulsion that each state must have an on outgoing edge for every input variable in:

Select the correct option

- ○ Transition Table
- ○ Finite Automata
- ○ Both Finite Automata and Transition Graph
- ● Transition Graph

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CS402:Grand Quiz

Quiz Start Time: 12:41 PM, 23 June 2021

Question # 11 of 30 (Start time: 12:45:20 PM, 23 June 2021)

Total Marks: 1

Let FA3 be an FA corresponding to FA1FA2, then the final state of FA3 must correspond to the final state of

Select the correct option

- (●) FA2 only

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- ( ) FA1 or FA2
- ( ) FA1 only
- ( ) FA1 and FA2

CS402:Grand Quiz

Quiz Start Time: 12:41 PM, 23 June 2021

Question # 12 of 30 (Start time: 12:45:44 PM, 23 June 2021)

Total Marks: 1

Kleene's Theorem Part I expresses the relationship between _____.

Select the correct option

- FA and RE
- FA and TG

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- RE and FA
- TG and RE

CS402:Grand Quiz

Quiz Start Time: 12:41 PM, 23 June 2021

Question # 13 of 30 (Start time: 12:46:14 PM, 23 June 2021)

Total Marks: 1

Two FAs are said to be equivalent, if they _____.

Select the correct option

- (●) accept same language

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- ( ) none of the given options
- ( ) accept null string
- ( ) accept different language

Question # 14 of 30 (Start time: 12:46:37 PM, 23 June 2021) **Total Marks:** 1

---------- can also help in proving Kleene Theorem III.

Select the correct option

- PDA
- Moore machine
- NFA

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- Melay machine

CS402:Grand Quiz

Quiz Start Time: 12:41 PM, 23 June 2021

Question # 15 of 30 (Start time: 12:47:00 PM, 23 June 2021)

Total Marks: 1

If A and B are regular languages, !(A' U B') is:

Select the correct option

- non regular
- may be regular
- regular

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- none of the mentioned

CS402:Grand Quiz

Quiz Start Time: 12:41 PM, 23 June 2021

Question # 16 of 30 (Start time: 12:47:29 PM, 23 June 2021)

Total Marks: 1

We cannot construct an NFA for the language of _____ defined over alphabet set {a,b}.

Select the correct option

- Palindromes

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- odd
- Even even
- Integers

CS402:Grand Quiz

Quiz Start Time: 12:41 PM, 23 June 2021

Question # 17 of 30 (Start time: 12:47:50 PM, 23 June 2021) **Total Marks: 1**

Closure of an FA is the same as _____ of an FA with itself except that the initial state of the required FA is a final state as well.

Select the correct option

- Sum
- Intersection
- Union
- Concatenation

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Saving...

CS402:Grand Quiz

Quiz Start Time: 12:41 PM, 23 June 2021

Question # 18 of 30 (Start time: 12:48:12 PM, 23 June 2021)

Total Marks: 1

Which of the following form correctly expressed the regular expression RR*?

Select the correct option

- R+R-
- R*
- R-
- R+

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CS402:Grand Quiz

Quiz Start Time: 12:41 PM, 23 June 2021

Question # 19 of 30 (Start time: 12:48:32 PM, 23 June 2021)

Total Marks: 1

Kleene Theorem III states that if the language can be expressed by RE then there exist -------- accepting the language.

Select the correct option

- (●) FA

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- ( ) Both TG and GTG
- ( ) GTG
- ( ) TG

CS402:Grand Quiz

Quiz Start Time: 12:41 PM, 23 June 2021

Question # 20 of 30 (Start time: 12:48:56 PM, 23 June 2021)

Total Marks: 1

If S = {a}, then S+ will be _____.

Select the correct option

- {a, aaa, aaaa, aaaaa,...}
- {a, aa, aaa, aaaa,...}
- {a, aaa, aaaaa, aaaaaaa,...}
- {aa, aaaa, aaaaaa, aaaaaaaa,...}

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CS402:Grand Quiz

Quiz Start Time: 12:41 PM, 23 June 2021

Question # 21 of 30 (Start time: 12:49:18 PM, 23 June 2021)

Total Marks: 1

Dead states are defined as:

Select the correct option

- It is non-distinguishable state
- A dead state is non final state of a DFA whose transition on every input symbol terminates on itself.
- No such state exist
- None of the given

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CS402:Grand Quiz

Quiz Start Time: 12:41 PM, 23 June 2021

Question # 22 of 30 (Start time: 12:49:43 PM, 23 June 2021)

Total Marks: 1

There _____ a language for which only FA can be built but not the RE.

Select the correct option

- may not be
- cannot be
- may be
- is

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Saving...

CS402:Grand Quiz

Quiz Start Time: 12:41 PM, 23 June 2021

Question # 23 of 30 (Start time: 12:50:07 PM, 23 June 2021) **Total Marks:** 1

Let S = {aa, bb}, then S* will have the _____ string.

Select the correct option

- aabbbaa
- Λ
- abba
- bbaab

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CS402:Grand Quiz

Quiz Start Time: 12:41 PM, 23 June 2021

Question # 24 of 30 (Start time: 12:50:33 PM, 23 June 2021) **Total Marks:** 1

Length of EVEN-EVEN language is _____.

Select the correct option

- Odd
- Such language doesn't exist
- Sometimes even & sometimes odd
- Even

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CS402:Grand Quiz

Quiz Start Time: 12:41 PM, 23 June 2021

Question # 25 of 30 (Start time: 12:51:00 PM, 23 June 2021)

Total Marks: 1

The minimum length of the strings(except null string) of a language that starts and ends in the same letters will be:

Select the correct option

- 2
- 1
- 3
- 4

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CS402:Grand Quiz

Quiz Start Time: 12:41 PM, 23 June 2021

Question # 26 of 30 (Start time: 12:51:26 PM, 23 June 2021)

Total Marks: 1

In Mealy machine the output depends on _____

Select the correct option

- (●) Present state and Present input

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- ( ) Nothing
- ( ) Only present state
- ( ) Type of input

CS402:Grand Quiz

Quiz Start Time: 12:41 PM, 23 June 2021

Question # 27 of 30 (Start time: 12:51:55 PM, 23 June 2021)

Total Marks: 1

Given S, Kleene star closure is denoted by _____.

Select the correct option

- S*
- S-
- S**
- S+

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CS402:Grand Quiz

Quiz Start Time: 12:41 PM, 23 June 2021

Question # 28 of 30 (Start time: 12:52:26 PM, 23 June 2021) **Total Marks:** 1

Which one of the following machine is represented as a pictorial representation with states and directed edges labeled by an input letter along with an output character?

Select the correct option

- Moore machine
- Mealy machine
- Finite state machine
- Deterministic finite state machine

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CS402:Grand Quiz

Quiz Start Time: 12:41 PM, 23 June 2021

Question # 29 of 30 (Start time: 12:52:52 PM, 23 June 2021)

Total Marks: 1

GTG for the expression (aa+aba)* may have minimum number of states:

Select the correct option

- 2
- 4
- 1
- 3

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CS402:Grand Quiz

Quiz Start Time: 12:41 PM, 23 June 2021

Question # 30 of 30 (Start time: 12:53:18 PM, 23 June 2021)

Total Marks: 1

FA and _____ are same except that _____ has unique symbol for each transition.

Select the correct option

- NFA,FA
- GTG,NFA

Not Confirm

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- FA,TG
- NFA,TG

[redacted] MUDASSAR NAZIR

Time Left 68 sec(s)

CS402:Grand Quiz

Quiz Start Time: 02:02 PM, 23 June 2021

Question # 1 of 30 (Start time: 02:02:21 PM, 23 June 2021) Total Marks: 1

A transition graph is converted into a(n) _____ in order to obtain regular expression.

Select the correct option

- FA
- GTG

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- NFA-^
- NFA

CS402:Grand Quiz

Quiz Start Time: 02:02 PM, 23 June 2021

Question # 2 of 30 (Start time: 02:03:00 PM, 23 June 2021)

Total Marks: 1

In proving Kleene Theorem II, circuits are reduced into:

Select the correct option

- both difference and asterisk
- sum
- asterisk

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- difference

Saving...

CS402:Grand Quiz

Quiz Start Time: 02:02 PM, 23 June 2021

Question # 3 of 30 (Start time: 02:03:23 PM, 23 June 2021)

Total Marks: 1

Which of the following is NOT true about the term alphabet?

Select the correct option

- It can be an empty set
- It is usually denoted by Greek letter sigma
- Strings are generated by concatenating its elements
- It is a finite set of symbols

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Saving

CS402:Grand Quiz

Quiz Start Time: 02:02 PM, 23 June 2021

Question # 4 of 30 (Start time: 02:03:54 PM, 23 June 2021)

Total Marks: 1

All possible combinations of strings of a language including null string is referred as:

Select the correct option

- Addition of a language with itself
- Multiplication of a language with itself
- Kleene star closure of a language

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- Concatenation of a language with itself

sec(s)

CS402:Grand Quiz

Quiz Start Time: 02:02 PM, 23 June 2021

Question # 5 of 30 (Start time: 02:04:41 PM, 23 June 2021)

Total Marks: 1

In concatenation, we include the initial state of FA2 automatically after the final state of FA1 because of:

Select the correct option

- We need just two initial states
- We need just one initial state
- Some part of the string may be accepted by FA2
- The strings of FA2 are accepted first before the strings of FA1

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CS402:Grand Quiz

Quiz Start Time: 02:02 PM, 23 June 2021

Question # 6 of 30 (Start time: 02:05:15 PM, 23 June 2021)

Total Marks: 1

If S = {a, b} then which of the following RE will generate all possible strings?

Select the correct option

- (ab + ba)*
- (ab)*
- a* + b*
- (a + b)*

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Question # 7 of 30 (Start time: 02:05:40 PM, 23 June 2021)

Total Marks: 1

Strings x,y,z belongs to Σ * such that both xz or yz ∈ L where L ⊆ Σ* are:

Select the correct option

- Indistinguishable

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- Undetermined
- Both distinguishable and indistinguishable
- Distinguishable

Question # 8 of 30 (Start time: 02:06:43 PM, 23 June 2021) **Total Marks:** 1

Let S = {a, bb, bab, baabb} be a set of strings, which one of the following will not be included in S*?

Select the correct option

- baabbabb
- bbbaabaabb
- bbaaabb
- baba

Question # 9 of 30 (Start time: 02:07:15 PM, 23 June 2021) **Total Marks:** 1

We cannot construct an NFA for the language of _____ defined over alphabet set {a,b}.

Select the correct option

- Integers
- odd
- Palindromes
- Even even

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CS402:Grand Quiz

Quiz Start Time: 02:02 PM, 23 June 2021

Question # 10 of 30 (Start time: 02:07:42 PM, 23 June 2021) **Total Marks:** 1

In Moore machine, if the length of input string is 9, then the length of output string will be:

Select the correct option

- 8
- 10

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- 7
- 9

CS402:Grand Quiz

Quiz Start Time: 02:02 PM, 23 June 2021

Question # 11 of 30 (Start time: 02:08:28 PM, 23 June 2021)

Total Marks: 1

Reverse of string "BcAbed" defined over Σ = {Ab, Bc, d,e} is _____.

Select the correct option

- deAbBc

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- debAcB
- debABc
- deAbcB

CS402:Grand Quiz

Quiz Start Time: 02:02 PM, 23 June 2021

Question # 12 of 30 (Start time: 02:09:02 PM, 23 June 2021)

Total Marks: 1

Every NFA can be considered to be a -------- as well, but the converse may not be true.

Select the correct option

- TG

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- PDA
- GTG
- FA

CS402:Grand Quiz

Quiz Start Time: 02:02 PM, 23 June 2021

Question # 13 of 30 (Start time: 02:09:36 PM, 23 June 2021)

Total Marks: 1

While finding RE corresponding to a TG, we connect the new start state with the old start state by _____ transition.

Select the correct option

- RE
- null
- a
- b

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Saving

CS402:Grand Quiz

Quiz Start Time: 02:02 PM, 23 June 2021

Question # 14 of 30 (Start time: 02:10:04 PM, 23 June 2021) **Total Marks: 1**

Which of the following is an utility of state elimination phenomenon?

Select the correct option

- All of the mentioned
- DFA to Regular Expression
- NFA to DFA
- DFA to NFA

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CS402:Grand Quiz

Quiz Start Time: 02:02 PM, 23 June 2021

Question # 15 of 30 (Start time: 02:10:39 PM, 23 June 2021)

Total Marks: 1

---------- state is not important in Moore machine.

Select the correct option

- Combination of
- Empty
- Initial

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- Final

CS402:Grand Quiz

Quiz Start Time: 02:02 PM, 23 June 2021

Question # 16 of 30 (Start time: 02:11:04 PM, 23 June 2021) **Total Marks:** 1

Decomposing a string into its valid units is referred as:

Select the correct option

- Decomposing
- Splitting
- Tokenizing

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- Dividing

CS402:Grand Quiz

Quiz Start Time: 02:02 PM, 23 June 2021

Question # 17 of 30 (Start time: 02:11:30 PM, 23 June 2021)

Total Marks: 1

GTG for the expression (aa+aba)* may have minimum number of states:

Select the correct option

1

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3

2

4

Saving...

CS402:Grand Quiz

Quiz Start Time: 02:02 PM, 23 June 2021

Question # 18 of 30 (Start time: 02:12:07 PM, 23 June 2021)

Total Marks: 1

If r1 is a regular expression then (r1)* is _____.

Select the correct option

- a finite automaton
- also a regular expression
- a non-deterministic finite automaton
- a generalized transition graph

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CS402:Grand Quiz

Quiz Start Time: 02:02 PM, 23 June 2021

Question # 19 of 30 (Start time: 02:12:37 PM, 23 June 2021)

Total Marks: 1

Which of the following diagrams expresses languages more simply?

Select the correct option

- NFA
- TG
- FA
- GTG

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Saving...

Question # 20 of 30 (Start time: 02:13:02 PM, 23 June 2021) **Total Marks:** 1

When ODD language is expressed by an FA, then it will have minimum _____ states.

Select the correct option

- two

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- three
- four
- one

Question # 21 of 30 (Start time: 02:13:37 PM, 23 June 2021) **Total Marks:** 1

In TG, there can be more than one _____.

Select the correct option

- final state
- start state and final state
- null state
- start state

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CS402:Grand Quiz

Quiz Start Time: 02:02 PM, 23 June 2021

Question # 22 of 30 (Start time: 02:14:13 PM, 23 June 2021)

Total Marks: 1

How many new states are introduced while developing NFA for the closure of an FA?

Select the correct option

- three
- one
- four
- two

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Saving...

CS402:Grand Quiz

Quiz Start Time: 02:02 PM, 23 June 2021

Question # 23 of 30 (Start time: 02:14:44 PM, 23 June 2021)

Total Marks: 1

In NFA having multiple transitions at certain state, FA can be built by introducing:

Select the correct option

- Combination of states

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- Initial state
- Empty state
- Final state

CS402:Grand Quiz

Quiz Start Time: 02:02 PM, 23 June 2021

Question # 24 of 30 (Start time: 02:15:23 PM, 23 June 2021)

Total Marks: 1

If two finite state machines are equivalent, they should have the same number of _____

Select the correct option

- states and edges
- states
- none of these
- edges

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Saving...

CS402:Grand Quiz

Quiz Start Time: 02:02 PM, 23 June 2021

Question # 25 of 30 (Start time: 02:15:48 PM, 23 June 2021)

Total Marks: 1

When even length language is expressed by an FA, then it will have minimum_____ states.

Select the correct option

- one
- four
- three
- two

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Saving...

CS402:Grand Quiz

Quiz Start Time: 02:02 PM, 23 June 2021

Question # 26 of 30 (Start time: 02:16:23 PM, 23 June 2021)

Total Marks: 1

What is false about the PALINDROME LANGUAGE?

Select the correct option

- FA can be build for it.

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- Every word is reverse of itself.
- None of the given option
- It is an infinite language.

CS402:Grand Quiz

Quiz Start Time: 02:02 PM, 23 June 2021

Question # 27 of 30 (Start time: 02:17:01 PM, 23 June 2021) **Total Marks:** 1

In regular expressions, the operator '*' stands for

Select the correct option

- Selection
- Iteration
- Addition
- Concatenation

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CS402:Grand Quiz

Quiz Start Time: 02:02 PM, 23 June 2021

Question # 28 of 30 (Start time: 02:17:56 PM, 23 June 2021)

Total Marks: 1

According to theory of automata there are _____ types of languages.

Select the correct option

- Four
- One
- Three
- Two

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CS402:Grand Quiz

Quiz Start Time: 02:02 PM, 23 June 2021

Question # 29 of 30 (Start time: 02:18:21 PM, 23 June 2021)

Total Marks: 1

We can create an equivalent _____ for a language for which we create an _____.

Select the correct option

- FA,NFA
- NFA,FA
- FA,GTG
- None of the given options

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CS402:Grand Quiz

Quiz Start Time: 02:02 PM, 23 June 2021

Question # 30 of 30 (Start time: 02:18:47 PM, 23 June 2021)

Total Marks: 1

FA is also called

Select the correct option

- NFA
- GTG
- DFA
- TG



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[redacted] MUHAMMAD FAIZAN

Time Left 74 sec(s)

CS402:Grand Quiz

Quiz Start Time: 01:19 PM, 23 June 2021

Question # 1 of 30 (Start time: 01:19:18 PM, 23 June 2021) Total Marks: 1

---------- state is not important in Moore machine.

Select the correct option

- Initial

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- Empty
- Combination of
- Final

Saving

Question # 2 of 30 (Start time: 01:19:44 PM, 23 June 2021) **Total Marks:** 1

Considering FA1 and FA2 having 2 states each. Now FA1+FA2 can have maximum _____ number of states.

Select the correct option

- 2
- None of the given options
- 3
- more than 3

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CS402:Grand Quiz

Quiz Start Time: 01:19 PM, 23 June 2021

Question # 3 of 30 (Start time: 01:20:08 PM, 23 June 2021)

Total Marks: 1

Regular languages are closed under the following operations.

Select the correct option

- Union only
- Concatenation, Closure only
- Union, Concatenation and Closure

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- Regular languages are not closed under any operation.

Saving...

Question # 4 of 30 (Start time: 01:20:30 PM, 23 June 2021) **Total Marks: 1**

Let FA1 accepts many strings and FA2 accepts no string, then FA1+FA2 will be equal to:

Select the correct option

- (FA2)*
- FA1
- FA2
- FA2-FA1

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CS402:Grand Quiz

Quiz Start Time: 01:19 PM, 23 June 2021

Question # 5 of 30 (Start time: 01:20:53 PM, 23 June 2021)

Total Marks: 1

How many new states are introduced while developing NFA for the closure of an FA?

Select the correct option

- four
- two
- three
- one

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CS402:Grand Quiz

Quiz Start Time: 01:19 PM, 23 June 2021

Question # 6 of 30 (Start time: 01:21:14 PM, 23 June 2021)

Total Marks: 1

Let FA1 has x number of states and FA2 has y number of states. Now FA1+FA2 can have maximum _____ number of states.

Select the correct option

- x/y
- None of the given options
- x+y
- x-y

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Saving...

CS402:Grand Quiz

Quiz Start Time: 01:19 PM, 23 June 2021

Question # 7 of 30 (Start time: 01:21:37 PM, 23 June 2021)

Total Marks: 1

While finding RE corresponding to TG, If TG has more than one start state then

Select the correct option

- Replace the old start state with final state
- Eliminate the old start state
- Introduce the new start state
- Replace the old final state with new start state

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CS402:Grand Quiz

Quiz Start Time: 01:19 PM, 23 June 2021

Question # 8 of 30 (Start time: 01:21:59 PM, 23 June 2021)

Total Marks: 1

If S = {a}, then S+ will be _____.

Select the correct option

- {aa, aaaa, aaaaaa, aaaaaaaa,...}
- {a, aa, aaa, aaaa,...}
- {a, aaa, aaaa, aaaaa,...}
- {a, aaa, aaaaa, aaaaaaa,...}

Solved By: Waqas Ahmed
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Saving...

Question # 9 of 30 (Start time: 01:22:22 PM, 23 June 2021) **Total Marks: 1**

Which of the following statement is true about GTG?

Select the correct option

- Transitions are based on input letters
- Transitions are based on regular expressions
- Transitions are based on specified substrings
- Transitions are based on alphabet set

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CS402:Grand Quiz

Quiz Start Time: 01:19 PM, 23 June 2021

Question # 10 of 30 (Start time: 01:22:43 PM, 23 June 2021)

Total Marks: 1

Which one of the following word is not accepted by the given regular expression?
(0+1)*(00+11)(0+1)*

Select the correct option

- 1010101

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- 1110011
- 0001100
- 0100010

CS402:Grand Quiz

Quiz Start Time: 01:19 PM, 23 June 2021

Question # 11 of 30 (Start time: 01:23:07 PM, 23 June 2021)

Total Marks: 1

While finding RE corresponding to a TG, we connect the new start state with the old start state by _____ transition.

Select the correct option

- a
- null

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- RE
- b

Saving...

Question # 12 of 30 (Start time: 01:23:31 PM, 23 June 2021) **Total Marks: 1**

Two machines are said to be equivalent if they print the _____ output string when the _____input string is run on them.

Select the correct option

- different, same
- same, different
- unique, different
- same, same

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Question # 13 of 30 (Start time: 01:23:50 PM, 23 June 2021)

Total Marks: 1

In concatenation, we include the initial state of FA2 automatically after the final state of FA1 because of:

Select the correct option

- The strings of FA2 are accepted first before the strings of FA1
- We need just two initial states
- Some part of the string may be accepted by FA2
- We need just one initial state

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CS402:Grand Quiz

Quiz Start Time: 01:19 PM, 23 June 2021

Question # 14 of 30 (Start time: 01:24:20 PM, 23 June 2021)

Total Marks: 1

Which of the following string belongs to the language of the regular expression (aa*b)*?

Select the correct option

- aaaaaa
- aabaab

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- abbbaa
- baabab

CS402:Grand Quiz

Quiz Start Time: 01:19 PM, 23 June 2021

Question # 15 of 30 (Start time: 01:25:05 PM, 23 June 2021) **Total Marks:** 1

Suppose we have an alphabet set having four symbols, how many transitions will be there on each state of a finite automaton for any language defined over the given alphabet?

Select the correct option

- 2
- 4 — Solved By: Waqas Ahmed
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- 1
- 3

Saving...

CS402:Grand Quiz

Quiz Start Time: 01:19 PM, 23 June 2021

Question # 16 of 30 (Start time: 01:25:35 PM, 23 June 2021)

Total Marks: 1

Reverse of string "BcAbed" defined over Σ = {Ab, Bc, d,e} is _____.

Select the correct option

- debAcB
- debABc
- deAbcB
- deAbBc

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CS402:Grand Quiz

Quiz Start Time: 01:19 PM, 23 June 2021

Question # 17 of 30 (Start time: 01:25:57 PM, 23 June 2021)

Total Marks: 1

1's complement for 01101 will be:

Select the correct option

- 10010

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- 00100
- 00000
- 10111

Saving

CS402:Grand Quiz

Quiz Start Time: 01:19 PM, 23 June 2021

Question # 18 of 30 (Start time: 01:26:23 PM, 23 June 2021)

Total Marks: 1

Dead states are defined as:

Select the correct option

- [x] A dead state is non final state of a DFA whose transition on every input symbol terminates on itself.
- [ ] No such state exist
- [ ] None of the given
- [ ] It is non-distinguishable state

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Saving...

CS402:Grand Quiz

Quiz Start Time: 01:19 PM, 23 June 2021

Question # 19 of 30 (Start time: 01:26:53 PM, 23 June 2021)

Total Marks: 1

A string will be accepted by an NFA if there exists _____ one successful path.

Select the correct option

- maximum
- none of the given options
- atmost
- atleast

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CS402:Grand Quiz

Quiz Start Time: 01:19 PM, 23 June 2021

Question # 20 of 30 (Start time: 01:27:16 PM, 23 June 2021)

Total Marks: 1

Which one of the following word is not accepted by the regular expression (a+b)*(aa)(a+b)*?

Select the correct option

- aa
- baaa
- aaaa
- bab

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Saving

CS402:Grand Quiz

Quiz Start Time: 01:19 PM, 23 June 2021

Question # 21 of 30 (Start time: 01:27:40 PM, 23 June 2021)

Total Marks: 1

A transition graph is converted into a(n) _____ in order to obtain regular expression.

Select the correct option

- GTG

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- NFA-^
- FA
- NFA

CS402:Grand Quiz

Quiz Start Time: 01:19 PM, 23 June 2021

Question # 22 of 30 (Start time: 01:28:07 PM, 23 June 2021)

Total Marks: 1

The language of all strings defined over alphabet set = {x, y} that ends with different letters will have the maximum length of:

Select the correct option

- infinite

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- 3
- 1
- 2

Saving...

CS402:Grand Quiz

Quiz Start Time: 01:19 PM, 23 June 2021

Question # 23 of 30 (Start time: 01:28:37 PM, 23 June 2021)

Total Marks: 1

If r1 and r2 are regular expressions then (r1 * r2) is _____ .

Select the correct option

- RE

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- FA
- GTG
- TG

CS402:Grand Quiz

Quiz Start Time: 01:19 PM, 23 June 2021

Question # 24 of 30 (Start time: 01:29:10 PM, 23 June 2021)

Total Marks: 1

Which of the following state is introduced while developing NFA for the closure of an FA?

Select the correct option

- [x] An initial state which should be final as well

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- [ ] An initial state with loop for all letters
- [ ] Final state
- [ ] Simply an initial state

Saving

CS402:Grand Quiz

Quiz Start Time: 01:19 PM, 23 June 2021

Question # 25 of 30 (Start time: 01:29:39 PM, 23 June 2021)

Total Marks: 1

In proving Kleene Theorem II, if a state has two incoming transition edges labelled by RE from the same state, then replace all the edges with a single transition edge labelled by ------- of corresponding RE.

Select the correct option

- Asterisk
- Concatenation
- Sum

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- Difference

Saving...

Question # 26 of 30 (Start time: 01:30:11 PM, 23 June 2021) **Total Marks: 1**

Kleene Theorem III states that if the language can be expressed by RE then there exist -------- accepting the language.

Select the correct option

- TG
- Both TG and GTG
- GTG
- FA

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CS402:Grand Quiz

Quiz Start Time: 01:19 PM, 23 June 2021

Question # 27 of 30 (Start time: 01:30:41 PM, 23 June 2021)

Total Marks: 1

The length of string "AbBAbcd" defined over Σ ={Ab,B,c,d} is _____.

Select the correct option

- five

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- three
- four
- six

Saving...

Question # 28 of 30 (Start time: 01:31:17 PM, 23 June 2021) **Total Marks:** 1

The aa(a+b)*bb is the RE of the language defined over Σ={a,b}.The language must _____.

Select the correct option

- have at least one ba
- have at least one aa and one bb
- have at least two ab
- have at least one abbb

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Question # 29 of 30 (Start time: 01:31:44 PM, 23 June 2021)

Total Marks: 1

Which of the following is not a step in elimination of states procedure?

Select the correct option

- Unifying all the final states into one using e-transitions
- Get the resulting regular expression by direct calculation
- Unify single transitions to multi transitions that contains union of input

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- Remove states until there is only starting and accepting states

CS402:Grand Quiz

Quiz Start Time: 01:19 PM, 23 June 2021

Question # 30 of 30 (Start time: 01:32:12 PM, 23 June 2021)

Total Marks: 1

Which one of the following word is not accepted by the given regular expression?
aa(a+b)*b

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Select the correct option

- aaabab
- abbaab
- aaaababb
- aabbabb

[redacted] REHANA KOUSAR

Time Left 71 sec(s)

CS402:Grand Quiz

Quiz Start Time: 12:13 PM, 23 June 2021

Question # 1 of 30 (Start time: 12:13:03 PM, 23 June 2021)

Total Marks: 1

FA and _____ are same except that _____ has unique symbol for each transition.

Select the correct option

- NFA,TG
- GTG,NFA
- FA,TG
- NFA,FA

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Question # 2 of 30 (Start time: 12:13:30 PM, 23 June 2021) **Total Marks: 1**

(aa + bb + (ab + ba)(aa + bb)*(ab + ba))* is the RE of _____.

Select the correct option

- FACTORIAL
- EVEN-EVEN
- ODD-ODD
- PALINDROME

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Question # 3 of 30 (Start time: 12:14:18 PM, 23 June 2021)

Total Marks: 1

In which of the following machine, the length of output string is 1 more than that of input string?

Select the correct option

- (●) Moore machine

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- ( ) Mealy machine
- ( ) Finite automaton with output
- ( ) Non-deterministic finite automaton

sec(s)

CS402:Grand Quiz

Quiz Start Time: 12:13 PM, 23 June 2021

Question # 4 of 30 (Start time: 12:14:46 PM, 23 June 2021) **Total Marks:** 1

If L1′ and L2′ are regular languages, then L1.L2 will be

Select the correct option

- non regular
- none of the mentioned
- may be regular
- regular

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CS402:Grand Quiz

Quiz Start Time: 12:13 PM, 23 June 2021

Question # 5 of 30 (Start time: 12:15:07 PM, 23 June 2021)

Total Marks: 1

Which one of the following word is not accepted by the given regular expression?
aa(a+b)*b

Select the correct option

- aaabab
- abbaab

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- aaaababb
- aabbabb

CS402:Grand Quiz

Quiz Start Time: 12:13 PM, 23 June 2021

Question # 6 of 30 (Start time: 12:15:34 PM, 23 June 2021)

Total Marks: 1

A regular language can be:

Select the correct option

- None of the given options
- irregular
- non-deterministic
- infinite

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CS402:Grand Quiz

Quiz Start Time: 12:13 PM, 23 June 2021

Question # 7 of 30 (Start time: 12:16:04 PM, 23 June 2021)

Total Marks: 1

How many states of a finite automaton will be final for accepting the only string 'abb'?

Select the correct option

- 2
- 1

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- 4
- 3

CS402:Grand Quiz

Quiz Start Time: 12:13 PM, 23 June 2021

Question # 8 of 30 (Start time: 12:16:32 PM, 23 June 2021)

Total Marks: 1

In GTG, there can be more than one:

Select the correct option

- start state
- final state
- null state
- start state and final state

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CS402:Grand Quiz

Quiz Start Time: 12:13 PM, 23 June 2021

Question # 9 of 30 (Start time: 12:16:58 PM, 23 June 2021)

Total Marks: 1

Consider the following RE:
a(a + b)b*
All of the following words are accepted except _____.

Select the correct option

- aab
- aa
- abb
- aba

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CS402:Grand Quiz

Quiz Start Time: 12:13 PM, 23 June 2021

Question # 10 of 30 (Start time: 12:17:28 PM, 23 June 2021)

Total Marks: 1

In Moore machine, if the length of input string is 9, then the length of output string will be:

Select the correct option

- 9
- 8
- 10
- 7

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CS402:Grand Quiz

Quiz Start Time: 12:13 PM, 23 June 2021

Question # 11 of 30 (Start time: 12:17:55 PM, 23 June 2021)

Total Marks: 1

If two RE's generate same language then these RE's are called _____.

Select the correct option

- Equal RE
- Same RE
- Equivalent RE
- Similar RE

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CS402:Grand Quiz

Quiz Start Time: 12:13 PM, 23 June 2021

Question # 12 of 30 (Start time: 12:18:23 PM, 23 June 2021)

Total Marks: 1

Every _____ is a _____ as well, but the converse may not be true.

Select the correct option

- TG, FA
- TG, GTG
- FA,GTG
- FA, TG

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Question # 13 of 30 (Start time: 12:18:50 PM, 23 June 2021) **Total Marks:** 1

Let's we have two regular expressions R1=(xx+yy) and R2=(x+ y). Which one of the following is the correct regular expression for the Union of R1 and R2?

Select the correct option

- (xx+yy)+(x+y)*
- (xx+yy)(x+y)
- (xx+yy)+(x+y)
- ((xx+yy)+(x+y))*

Solved By: Waqas Ahmed
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Question # 14 of 30 (Start time: 12:19:51 PM, 23 June 2021) **Total Marks:** 1

The length of string "AbBAbcd" defined over Σ ={Ab,B,c,d} is _____.

Select the correct option

- (●) five

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- ( ) four
- ( ) three
- ( ) six

CS402:Grand Quiz

Quiz Start Time: 12:13 PM, 23 June 2021

Question # 15 of 30 (Start time: 12:20:30 PM, 23 June 2021)

Total Marks: 1

The language of all strings defined over alphabet set = {a, b} containing 'bbb' will have the minimum string with length of:

Select the correct option

- (●) 3

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- ( ) 2
- ( ) 1
- ( ) 4

Question # 16 of 30 (Start time: 12:21:49 PM, 23 June 2021) **Total Marks:** 1

If S = {ab, bb} then S* will not contain _____.

Select the correct option

- bbba

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- ababbb
- bbbbab
- abbbab

Saving

CS402:Grand Quiz

Quiz Start Time: 12:13 PM, 23 June 2021

Question # 17 of 30 (Start time: 12:22:26 PM, 23 June 2021)

Total Marks: 1

Kleene's Theorem Part III expresses the relationship between _____.

Select the correct option

- TG and RE
- FA and TG
- FA and RE
- RE and FA

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sec(s)

CS402:Grand Quiz **Quiz Start Time: 12:13 PM, 23 June 2021**

Question # 18 of 30 (Start time: 12:23:07 PM, 23 June 2021) **Total Marks:** 1

There _____ be a unique path for each valid string (called a word) in NFA.

Select the correct option

- must
- will
- may not
- should not

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CS402:Grand Quiz

Quiz Start Time: 12:13 PM, 23 June 2021

Question # 19 of 30 (Start time: 12:24:05 PM, 23 June 2021)

Total Marks: 1

If an FA has 3 states and 2 letters in the alphabet set, then it will have total _____ number of transitions.

Select the correct option

- (●) 6

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- ( ) 5
- ( ) 7
- ( ) 4

CS402:Grand Quiz

Quiz Start Time: 12:13 PM, 23 June 2021

Question # 20 of 30 (Start time: 12:24:43 PM, 23 June 2021) **Total Marks:** 1

Suppose we have an alphabet set having four symbols, how many transitions will be there on each state of a finite automaton for any language defined over the given alphabet?

Select the correct option

- (●) 4

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- ( ) 1
- ( ) 2
- ( ) 3

CS402:Grand Quiz

Quiz Start Time: 12:13 PM, 23 June 2021

Question # 21 of 30 (Start time: 12:25:14 PM, 23 June 2021)

Total Marks: 1

Which one of the following is a correct word produced by the RE (a*b*)ab?

Select the correct option

- aaaa
- Null
- abab
- bbbb

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Question # 22 of 30 (Start time: 12:25:47 PM, 23 June 2021) **Total Marks: 1**

Which of the following statement is NOT true about TG?

Select the correct option

- ○ There may exist no path for certain string
- ○ There may be no final state
- ● There exists exactly one path for certain string
- ○ There may exist more than one paths for certain string

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CS402:Grand Quiz

Quiz Start Time: 12:13 PM, 23 June 2021

Question # 23 of 30 (Start time: 12:26:14 PM, 23 June 2021)

Total Marks: 1

Substrings as input letters can be specified on edges in:

Select the correct option

- (●) TG

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- ( ) PDA
- ( ) FA
- ( ) NFA

Question # 24 of 30 (Start time: 12:26:41 PM, 23 June 2021) **Total Marks: 1**

a(a+b)*b + b(a+b)*a is the regular expression of language defined over Σ={a,b} that is _____.

Select the correct option

- starting with a and ending in a
- starting with a and ending in b or starting with b and ending in a
- starting with b and ending in a
- starting with a and ending in b

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Question # 25 of 30 (Start time: 12:27:08 PM, 23 June 2021) **Total Marks:** 1

Which one of the following is the RE for the language defined over Σ= {a, b} having all the words starting with a?

Select the correct option

- a*(a + b)
- aa(a + b)+
- a(a + b)*
- (a + b)*

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CS402:Grand Quiz

Quiz Start Time: 12:13 PM, 23 June 2021

Question # 26 of 30 (Start time: 12:27:33 PM, 23 June 2021)

Total Marks: 1

The language of all strings defined over alphabet set = {x, y} that ends with same letters will have the maximum length of:

Select the correct option

- 2
- 3
- infinite

Solved By: Waqas Ahmed
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- 1

CS402:Grand Quiz

Quiz Start Time: 12:13 PM, 23 June 2021

Question # 27 of 30 (Start time: 12:27:57 PM, 23 June 2021)

Total Marks: 1

Let S = {aa, bb}, then S* will have the _____ string.

Select the correct option

- aabbbaa
- bbaab
- abba
- Λ

Solved By: Waqas Ahmed
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CS402:Grand Quiz

Quiz Start Time: 12:13 PM, 23 June 2021

Question # 28 of 30 (Start time: 12:28:20 PM, 23 June 2021)

Total Marks: 1

There may be more than one transition for a certain letter on a state in:

Select the correct option

- Moore Machine
- Transition Table
- Finite Automata
- Non-Deterministic Finite Automata

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Question # 30 of 30 (Start time: 12:29:15 PM, 23 June 2021)

Melay machine to increase the output string in magnitude by 1 is called:

Select the correct option

- Converting machine
- Decrementing machine
- Incrementing machine
- Complementing machine

[redacted] MUHAMMAD SULEMAN SHAFQAT

Time Left 69 sec(s)

CS402:Grand Quiz

Quiz Start Time: 12:55 PM, 23 June 2021

Question # 1 of 30 (Start time: 12:55:46 PM, 23 June 2021) Total Marks: 1

Which of the following is NOT true about the term alphabet?

Select the correct option

- (●) It can be an empty set

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- ( ) Strings are generated by concatenating its elements
- ( ) It is a finite set of symbols
- ( ) It is usually denoted by Greek letter sigma

Question # 2 of 30 (Start time: 12:56:18 PM, 23 June 2021) **Total Marks:** 1

Consider the following Regular Expression (RE):
a(a+b)*b+b(a+b)a
All of the following words are accepted except _____.

Select the correct option

- bba
- bab
- abaab
- aab

Solved By: Waqas Ahmed
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CS402:Grand Quiz

Quiz Start Time: 12:55 PM, 23 June 2021

Question # 3 of 30 (Start time: 12:57:09 PM, 23 June 2021) **Total Marks:** 1

Let's we have two regular expressions R1=(xx+yy) and R2=(x+ y). Which one of the following is the correct regular expression for the Union of R1 and R2?

Select the correct option

- (xx+yy)(x+y)
- (xx+yy)+(x+y)*
- (xx+yy)+(x+y)
- ((xx+yy)+(x+y))*

Solved By: Waqas Ahmed
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CS402:Grand Quiz

Quiz Start Time: 12:55 PM, 23 June 2021

Question # 4 of 30 (Start time: 12:57:41 PM, 23 June 2021)

Total Marks: 1

Every _____ is a _____ as well, but the converse may not be true.

Select the correct option

- FA, TG

Solved By: Waqas Ahmed
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- TG, GTG
- FA,GTG
- TG, FA

Saving

CS402:Grand Quiz

Quiz Start Time: 12:55 PM, 23 June 2021

Question # 5 of 30 (Start time: 12:58:08 PM, 23 June 2021)

Total Marks: 1

While developing NFA for the union of FA1 and FA2, if there is a loop of 'a' at the initial state of FA1 then the new initial state will have a transition for 'a' that goes straight to:

Select the correct option

- the final state of FA1
- the initial state of FA1*FA1
- the initial state of FA2
- the initial state of FA1

Solved By: Waqas Ahmed
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CS402:Grand Quiz

Quiz Start Time: 12:55 PM, 23 June 2021

Question # 6 of 30 (Start time: 12:58:38 PM, 23 June 2021)

Total Marks: 1

Every NFA can be considered to be a -------- as well, but the converse may not be true.

Select the correct option

- PDA
- FA
- TG
- GTG

Solved By: Waqas Ahmed
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CS402:Grand Quiz

Quiz Start Time: 12:55 PM, 23 June 2021

Question # 7 of 30 (Start time: 12:59:03 PM, 23 June 2021)

Total Marks: 1

In NFA having no transition at certain state, FA can be built by introducing:

Select the correct option

- Final state
- Initial state
- Empty state

Solved By: Waqas Ahmed
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- Combination of states

CS402:Grand Quiz

Quiz Start Time: 12:55 PM, 23 June 2021

Question # 8 of 30 (Start time: 12:59:29 PM, 23 June 2021) **Total Marks:** 1

Consider we have languages L7 and L6. Which of the following represents their concatenation?

Select the correct option

- L7L6

Solved By: Waqas Ahmed
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- L6*L7
- L7+L6
- L7/L6

Saving

CS402:Grand Quiz

Quiz Start Time: 12:55 PM, 23 June 2021

Question # 9 of 30 (Start time: 12:59:54 PM, 23 June 2021)

Total Marks: 1

In Moore machine, output is produced over the change of:

Select the correct option

- Transitions and states
- None of the mentioned
- Transitions
- States

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CS402:Grand Quiz

Quiz Start Time: 12:55 PM, 23 June 2021

Question # 10 of 30 (Start time: 01:00:24 PM, 23 June 2021)

Total Marks: 1

While finding RE corresponding to a TG, we connect the new start state with the old start state by _____ transition.

Select the correct option

- null
- RE
- a
- b

Solved By: Waqas Ahmed
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CS402:Grand Quiz

Quiz Start Time: 12:55 PM, 23 June 2021

Question # 11 of 30 (Start time: 01:00:48 PM, 23 June 2021)

Total Marks: 1

Consider the following RE:
a(a + b)b*
All of the following words are accepted except _____.

Select the correct option

- aa
- aab
- abb
- aba

Solved By: Waqas Ahmed
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Saving...

CS402:Grand Quiz

Quiz Start Time: 12:55 PM, 23 June 2021

Question # 12 of 30 (Start time: 01:01:09 PM, 23 June 2021)

Total Marks: 1

There _____ be a unique path for each valid string (called a word) in NFA.

Select the correct option

- should not
- must
- will
- may not

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Saving

CS402:Grand Quiz

Quiz Start Time: 12:55 PM, 23 June 2021

Question # 13 of 30 (Start time: 01:01:34 PM, 23 June 2021)

Total Marks: 1

There may be more than one transition for a certain letter on a state in:

Select the correct option

- Finite Automata
- Moore Machine
- Non-Deterministic Finite Automata

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- Transition Table

CS402:Grand Quiz

Quiz Start Time: 12:55 PM, 23 June 2021

Question # 14 of 30 (Start time: 01:02:00 PM, 23 June 2021)

Total Marks: 1

Kleene Theorem III states that if the language can be expressed by RE then there exist -------- accepting the language.

Select the correct option

- FA

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- GTG
- Both TG and GTG
- TG

CS402:Grand Quiz

Quiz Start Time: 12:55 PM, 23 June 2021

Question # 15 of 30 (Start time: 01:02:34 PM, 23 June 2021)

Total Marks: 1

What is false about the PALINDROME LANGUAGE?

Select the correct option

- It is an infinite language.
- FA can be build for it.
- Every word is reverse of itself.
- None of the given option

Solved By: Waqas Ahmed
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CS402:Grand Quiz

Quiz Start Time: 12:55 PM, 23 June 2021

Question # 16 of 30 (Start time: 01:03:01 PM, 23 June 2021)

Total Marks: 1

Let FA3 be an FA corresponding to FA1FA2, then the initial state of FA3 must correspond to the initial state of

Select the correct option

- FA1 or FA2
- FA1 only
- FA1 and FA2
- FA2 only

Solved By: Waqas Ahmed
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Question # 17 of 30 (Start time: 01:03:32 PM, 23 June 2021) **Total Marks: 1**

All possible combinations of strings of a language including null string is referred as:

Select the correct option

- Multiplication of a language with itself
- Concatenation of a language with itself
- Addition of a language with itself
- Kleene star closure of a language

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CS402:Grand Quiz

Quiz Start Time: 12:55 PM, 23 June 2021

Question # 18 of 30 (Start time: 01:04:00 PM, 23 June 2021)

Total Marks: 1

In Moore machine, if the length of input string is 9, then the length of output string will be:

Select the correct option

- 9
- 7
- 10
- 8

Solved By: Waqas Ahmed
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CS402:Grand Quiz

Quiz Start Time: 12:55 PM, 23 June 2021

Question # 19 of 30 (Start time: 01:04:25 PM, 23 June 2021)

Total Marks: 1

If we have a finite language and the number of states in the FA is n then the maximum number of letters in the each word of the language that will be accepted by the given FA will be:

Select the correct option

- n+1
- n
- n-1
- 1

Solved By: Waqas Ahmed
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CS402:Grand Quiz

Quiz Start Time: 12:55 PM, 23 June 2021

Question # 20 of 30 (Start time: 01:04:48 PM, 23 June 2021)

Total Marks: 1

Null strings can be specified on edges in:

Select the correct option

- Finite Automata
- Transition Graph
- Non-Deterministic Finite Automata
- Melay Machine

Solved By: Waqas Ahmed
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Question # 21 of 30 (Start time: 01:05:15 PM, 23 June 2021)

Total Marks: 1

Which of the following state is introduced while developing NFA for the closure of an FA?

Select the correct option

- An initial state with loop for all letters
- Simply an initial state
- Final state
- An initial state which should be final as well

Solved By: Waqas Ahmed
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CS402:Grand Quiz

Quiz Start Time: 12:55 PM, 23 June 2021

Question # 22 of 30 (Start time: 01:05:38 PM, 23 June 2021)

Total Marks: 1

According to theory of automata there are _____ types of languages.

Select the correct option

- Three
- One
- Four
- Two

Solved By: Waqas Ahmed
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CS402:Grand Quiz

Quiz Start Time: 12:55 PM, 23 June 2021

Question # 23 of 30 (Start time: 01:06:02 PM, 23 June 2021)

Total Marks: 1

Which of the following doesn't accept any language?

Select the correct option

- Deterministic finite state machine
- Finite state machine
- Regular expression
- Moore machine

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CS402:Grand Quiz

Quiz Start Time: 12:55 PM, 23 June 2021

Question # 24 of 30 (Start time: 01:06:27 PM, 23 June 2021)

Total Marks: 1

Substrings as input letters can be specified on edges in:

Select the correct option

- TG

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- NFA
- PDA
- FA

Saving

CS402:Grand Quiz

Quiz Start Time: 12:55 PM, 23 June 2021

Question # 25 of 30 (Start time: 01:06:59 PM, 23 June 2021)

Total Marks: 1

The language of all strings defined over alphabet set = {a, b} containing 'bbb' will have the minimum string with length of:

Select the correct option

- 4
- 1
- 2
- 3

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Saving...

CS402:Grand Quiz

Quiz Start Time: 12:55 PM, 23 June 2021

Question # 26 of 30 (Start time: 01:07:21 PM, 23 June 2021)

Total Marks: 1

The language of all strings defined over alphabet set = {x, y} that ends with same letters will have the maximum length of:

Select the correct option

- 1
- 3
- 2
- infinite

Solved By: Waqas Ahmed
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sec(s)

CS402:Grand Quiz

Quiz Start Time: 12:55 PM, 23 June 2021

Question # 27 of 30 (Start time: 01:07:46 PM, 23 June 2021) **Total Marks:** 1

In TG:
Statement 1: There may exist more than one path for a certain string
Statement 2: There may not exist any path for a certain string
Which of the above statement(s) is/are true or false?

Select the correct option

- 2 true only
- Both 1 and 2 are true
- Both 1 and 2 are false
- 1 true only

Solved By: Waqas Ahmed
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CS402:Grand Quiz

Quiz Start Time: 12:55 PM, 23 June 2021

Question # 28 of 30 (Start time: 01:08:19 PM, 23 June 2021)

Total Marks: 1

Let S = {a, bb, bab, baabb} be a set of strings, which one of the following will not be included in S*?

Select the correct option

- baba
- baabbabb
- bbaaabb
- bbbaabaabb

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CS402:Grand Quiz

Quiz Start Time: 12:55 PM, 23 June 2021

Question # 29 of 30 (Start time: 01:08:44 PM, 23 June 2021)

Total Marks: 1

Reverse of string "BcAbed" defined over Σ = {Ab, Bc, d,e} is _____.

Select the correct option

- deAbBc

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- deAbcB
- debABc
- debAcB

CS402:Grand Quiz

Quiz Start Time: 12:55 PM, 23 June 2021

Question # 30 of 30 (Start time: 01:09:09 PM, 23 June 2021)

Total Marks: 1

Which one of the following word is not accepted by the given regular expression?
aa(a+b)*b

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Select the correct option

- aaabab
- abbaab
- aaaababb
- aabbabb

CS402:Grand Quiz

Quiz Start Time: 05:06 PM, 23 June 2021

Question # 1 of 30 (Start time: 05:06:08 PM, 23 June 2021) **Total Marks: 1**

Consider we have languages L7 and L6. Which of the following represents their concatenation?

Select the correct option

- L6*L7
- L7+L6
- L7L6

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- L7/L6

CS402:Grand Quiz

Quiz Start Time: 05:06 PM, 23 June 2021

Question # 2 of 30 (Start time: 05:06:38 PM, 23 June 2021)

Total Marks: 1

If S = {a}, then S+ will be _____.

Select the correct option

- {a, aa, aaa, aaaa,...}

Solved By: Waqas Ahmed
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- {a, aaa, aaaa, aaaaa,...}
- {a, aaa, aaaaa, aaaaaaa,...}
- {aa, aaaa, aaaaaa, aaaaaaaa,...}

Saving...

CS402:Grand Quiz

Quiz Start Time: 05:06 PM, 23 June 2021

Question # 3 of 30 (Start time: 05:06:55 PM, 23 June 2021) **Total Marks:** 1

In proving Kleene Theorem II, if three states are connected then middle state is removed by connecting first and third state and writing corresponding RE in:

Select the correct option

- ○ Difference
- ● Concatenation
- ○ Sum
- ○ Asterisk

Solved By: Waqas Ahmed
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Saving

CS402:Grand Quiz

Quiz Start Time: 05:06 PM, 23 June 2021

Question # 4 of 30 (Start time: 05:07:08 PM, 23 June 2021)

Total Marks: 1

While finding RE corresponding to a TG, we connect the new start state with the old start state by _____ transition.

Select the correct option

- RE
- null

Solved By: Waqas Ahmed
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- b
- a

Question # 5 of 30 (Start time: 05:07:22 PM, 23 June 2021) **Total Marks:** 1

Which of the following machine has only one initial state and no final state?

Select the correct option

- (●) Moore machine

Solved By: Waqas Ahmed
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- ( ) Non deterministic finite state machine
- ( ) Finite state machine
- ( ) Deterministic finite state machine

CS402:Grand Quiz

Quiz Start Time: 05:06 PM, 23 June 2021

Question # 6 of 30 (Start time: 05:07:42 PM, 23 June 2021) **Total Marks:** 1

The language of all strings defined over alphabet set = {x, y} that ends with same letters will have the maximum length of:

Select the correct option

- 2
- infinite

Solved By: Waqas Ahmed
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- 3
- 1

CS402:Grand Quiz

Quiz Start Time: 05:06 PM, 23 June 2021

Question # 7 of 30 (Start time: 05:08:10 PM, 23 June 2021) **Total Marks:** 1

How many states of a finite automaton will be final for accepting L = {^, b, bb, bbb}?

Select the correct option

- 1
- 2
- 3
- 4

Solved By: Waqas Ahmed
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CS402:Grand Quiz

Quiz Start Time: 05:06 PM, 23 June 2021

Question # 8 of 30 (Start time: 05:08:27 PM, 23 June 2021)

Total Marks: 1

When ODD language is expressed by an FA, then it will have minimum _____ states.

Select the correct option

- three
- four
- two

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- one

CS402:Grand Quiz

Quiz Start Time: 05:06 PM, 23 June 2021

Question # 9 of 30 (Start time: 05:08:59 PM, 23 June 2021) **Total Marks: 1**

FA is also called

Select the correct option

- NFA
- DFA

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- GTG
- TG

CS402:Grand Quiz

Quiz Start Time: 05:06 PM, 23 June 2021

Question # 10 of 30 (Start time: 05:09:34 PM, 23 June 2021) **Total Marks:** 1

Which of the following is an utility of state elimination phenomenon?

Select the correct option

- DFA to NFA
- NFA to DFA
- DFA to Regular Expression

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- All of the mentioned

Question # 11 of 30 (Start time: 05:09:49 PM, 23 June 2021) **Total Marks:** 1

While developing NFA for the union of FA1 and FA2, there will be _____ transition/transitions for both 'a' and 'b' on the new initial state.

Select the correct option

- Only one
- Single
- Only three
- Multiple

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CS402:Grand Quiz

Quiz Start Time: 05:06 PM, 23 June 2021

Question # 12 of 30 (Start time: 05:10:04 PM, 23 June 2021)

Total Marks: 1

Non-Determinism is present in:

Select the correct option

- FA
- Both TG and GTG
- TG
- GTG

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Question # 13 of 30 (Start time: 05:10:19 PM, 23 June 2021) **Total Marks:** 1

In FA, initial state can be represented by:

Select the correct option

- leave the state empty
- drawing '+' sign in that state
- drawing a circle in that state
- drawing an arrow head before that state

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CS402:Grand Quiz

Quiz Start Time: 05:06 PM, 23 June 2021

Question # 14 of 30 (Start time: 05:10:35 PM, 23 June 2021) **Total Marks:** 1

Formal is also known as _____.

Select the correct option

- ○ None of these
- ○ Semantic language
- ● Syntactic language
- ○ Informal language

Solved By: Waqas Ahmed
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CS402:Grand Quiz

Quiz Start Time: 05:06 PM, 23 June 2021

Question # 15 of 30 (Start time: 05:10:48 PM, 23 June 2021) **Total Marks:** 1

Keeping in view the language of all strings ending with 'a', for which symbol we will take a loop on the final state of its transition diagram?

Select the correct option

- b
- d
- a

Solved By: Waqas Ahmed
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- c

CS402:Grand Quiz

Quiz Start Time: 05:06 PM, 23 June 2021

Question # 16 of 30 (Start time: 05:11:25 PM, 23 June 2021)

Total Marks: 1

Alphabet S = {a, bc, cc} has _____ number of letters.

Select the correct option

- Two
- One
- Four
- Three

Solved By: Waqas Ahmed
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Question # 17 of 30 (Start time: 05:12:01 PM, 23 June 2021) **Total Marks: 1**

If r1 is a regular expression then (r1)* is _____.

Select the correct option

- (●) also a regular expression

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- ( ) a finite automaton
- ( ) a non-deterministic finite automaton
- ( ) a generalized transition graph

CS402:Grand Quiz

Quiz Start Time: 05:06 PM, 23 June 2021

Question # 18 of 30 (Start time: 05:12:18 PM, 23 June 2021)

Total Marks: 1

Which of the following is free of non-determinism?

Select the correct option

- NFA-^
- TG
- FA

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- NFA

Saving...

CS402:Grand Quiz

Quiz Start Time: 05:06 PM, 23 June 2021

Question # 19 of 30 (Start time: 05:12:30 PM, 23 June 2021) **Total Marks: 1**

In Moore machine the output depends on

Select the correct option

- Present state and present input
- Type of input
- Only present state

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- Nothing

CS402:Grand Quiz

Quiz Start Time: 05:06 PM, 23 June 2021

Question # 20 of 30 (Start time: 05:12:43 PM, 23 June 2021) **Total Marks: 1**

What is false about the PALINDROME LANGUAGE?

Select the correct option

- None of the given option
- Every word is reverse of itself.
- FA can be build for it.
- It is an infinite language.

Solved By: Waqas Ahmed
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CS402:Grand Quiz

Quiz Start Time: 05:06 PM, 23 June 2021

Question # 21 of 30 (Start time: 05:13:35 PM, 23 June 2021) **Total Marks:** 1

Which one of the following machine is represented as a pictorial representation with states and directed edges labeled by an input letter along with an output character?

Select the correct option

- ○ Finite state machine
- ○ Moore machine
- ● Mealy machine
- ○ Deterministic finite state machine

Solved By: Waqas Ahmed
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Question # 22 of 30 (Start time: 05:13:51 PM, 23 June 2021) **Total Marks:** 1

If an alphabet has "2" number of letters, then total number of strings of length "3" will be _____.

Select the correct option

- 5
- 6
- 8 Solved By: Waqas Ahmed
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- 9

Question # 23 of 30 (Start time: 05:14:03 PM, 23 June 2021) **Total Marks: 1**

Let L be the language of all strings, defined over Σ = {0,1}, ending in 10. Which of the following strings are indistinguishable with respect to L with z being 0?

Select the correct option

- ○ 100, 101
- ○ 010, 101
- ○ 110, 101
- ● 111, 101

Solved By: Waqas Ahmed
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Question # 24 of 30 (Start time: 05:14:47 PM, 23 June 2021)

Total Marks: 1

The FA can be drawn for the regular expression $(a+b)^*$ with minimum _____ state(s).

Select the correct option

- 3
- 2
- 1

Solved By: Waqas Ahmed
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- 4

CS402:Grand Quiz

Quiz Start Time: 05:06 PM, 23 June 2021

Question # 25 of 30 (Start time: 05:15:20 PM, 23 June 2021)

Total Marks: 1

Which of the following diagrams expresses languages more simply?

Select the correct option

- NFA
- GTG

Solved By: Waqas Ahmed
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- TG
- FA

Question # 26 of 30 (Start time: 05:15:38 PM, 23 June 2021) **Total Marks:** 1

In Moore machine, output is produced over the change of:

Select the correct option

- (●) States

Solved By: Waqas Ahmed
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- ( ) Transitions
- ( ) None of the mentioned
- ( ) Transitions and states

CS402:Grand Quiz

Quiz Start Time: 05:06 PM, 23 June 2021

Question # 27 of 30 (Start time: 05:15:54 PM, 23 June 2021) **Total Marks:** 1

In the context of make NFA for the concatenation of FA1 and FA2 (FA2 accepting null string), which of the following option is correct?

Select the correct option

- (●) Final states in both FAs

Solved By: Waqas Ahmed
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- ( ) Initial states in both FAs
- ( ) FA2 having initial state only
- ( ) FA2 having final state only

Saving...

CS402:Grand Quiz

Quiz Start Time: 05:06 PM, 23 June 2021

Question # 28 of 30 (Start time: 05:16:37 PM, 23 June 2021) **Total Marks: 1**

Which one of the following word is not accepted by the given regular expression?
$(a+b)^*bba(a+b)^*$

Select the correct option

- babbba
- bbabbbbaba
- aaababab
- abbbbaa

Solved By: Waqas Ahmed
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Question # 29 of 30 (Start time: 05:16:53 PM, 23 June 2021) **Total Marks: 1**

Introducing new start state in case of multiple start states is the step no. _____ of proving Kleene's theorem part II.

Select the correct option

- 2
- 4
- 1

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- 3

CS402:Grand Quiz

Question # 30 of 30 (Start time: 05:17:16 PM, 23 June 2021)

Substrings as input letters can be specified on edges in:

Select the correct option

- ○ NFA
- ● TG
  Solved By: Waqas Ahmed
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- ○ PDA
- ○ FA

[redacted]: MUHAMMAD KHALEEQ

Time Left 75 sec(s)

CS402:Grand Quiz

Quiz Start Time: 01:44 PM, 23 June 2021

Question # 1 of 30 (Start time: 01:44:35 PM, 23 June 2021) Total Marks: 1

Mealy machine is equivalent to Moore machine, if we:

Select the correct option

- Ignore the extra input character inserted by the Moore machine
- Ignore the extra input character inserted by the Mealy machine
- Ignore the extra character printed by the Mealy machine
- Ignore the extra character printed by the Moore machine

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CS402:Grand Quiz

Quiz Start Time: 01:44 PM, 23 June 2021

Question # 2 of 30 (Start time: 01:45:01 PM, 23 June 2021)

Total Marks: 1

In which of the following machine, the length of output string is 1 more than that of input string?

Select the correct option

- (●) Moore machine

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- ( ) Mealy machine
- ( ) Finite automaton with output
- ( ) Non-deterministic finite automaton

CS402:Grand Quiz

Quiz Start Time: 01:44 PM, 23 June 2021

Question # 3 of 30 (Start time: 01:45:28 PM, 23 June 2021)

Total Marks: 1

In NFA having no transition at certain state, FA can be built by introducing:

Select the correct option

- Empty state

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- Final state
- Combination of states
- Initial state

CS402:Grand Quiz

Quiz Start Time: 01:44 PM, 23 June 2021

Question # 4 of 30 (Start time: 01:45:56 PM, 23 June 2021) **Total Marks:** 1

A loop at a state is supposed to be ---------- transition while converting Moore machine into an equivalent Melay machine.

Select the correct option

- Both incoming and outgoing
- Outgoing
- Complex
- Incoming

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CS402:Grand Quiz

Quiz Start Time: 01:44 PM, 23 June 2021

Question # 5 of 30 (Start time: 01:46:35 PM, 23 June 2021) **Total Marks:** 1

Suppose we have FA3 (which is equal to FA1 + FA2), then the final state of FA3 will be declared final if:

Select the correct option

- It corresponds to final states of both FA1 and FA2
- It corresponds to any of the final states in FA1 or FA2
- It corresponds to final states of FA1 only
- It corresponds to final states of FA2 only

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CS402:Grand Quiz

Quiz Start Time: 01:44 PM, 23 June 2021

Question # 6 of 30 (Start time: 01:47:06 PM, 23 June 2021)

Total Marks: 1

Which one of the following is the RE for the language defined over Σ= {a, b} having all the words starting with a?

Select the correct option

- (a + b)*
- a*(a + b)
- a(a + b)*
- aa(a + b)+

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Quiz Start Time: 01:44 PM, 23 June 2021

Question # 7 of 30 (Start time: 01:47:30 PM, 23 June 2021) **Total Marks:** 1

If r1 and r2 are regular expressions then (r1 * r2) is _____ .

Select the correct option

- GTG
- RE

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- TG
- FA

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Quiz Start Time: 01:44 PM, 23 June 2021

Question # 8 of 30 (Start time: 01:48:01 PM, 23 June 2021)

Total Marks: 1

While developing NFA for the union of FA1 and FA2, there will be _____ transition/transitions for both 'a' and 'b' on the new initial state.

Select the correct option

- Multiple

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- Only one
- Single
- Only three

Saving

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Quiz Start Time: 01:44 PM, 23 June 2021

Question # 9 of 30 (Start time: 01:49:05 PM, 23 June 2021)

Total Marks: 1

If S = {a, b} then which of the following RE will generate all possible strings?

Select the correct option

- $a^* + b^*$
- $(a + b)^*$

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- $(ab)^*$
- $(ab + ba)^*$

Saving...

CS402:Grand Quiz

Quiz Start Time: 01:44 PM, 23 June 2021

Question # 10 of 30 (Start time: 01:49:33 PM, 23 June 2021)

Total Marks: 1

If "r1" and "r2" are regular expressions, then which of the following is not a regular expression?

Select the correct option

- r1 - r2

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- r1 + r2
- r1*
- r1 r2

Saving...

CS402:Grand Quiz

Quiz Start Time: 01:44 PM, 23 June 2021

Question # 11 of 30 (Start time: 01:50:04 PM, 23 June 2021) **Total Marks:** 1

In concatenation, we include the initial state of FA2 automatically after the final state of FA1 because of:

Select the correct option

- We need just one initial state
- The strings of FA2 are accepted first before the strings of FA1
- Some part of the string may be accepted by FA2
- We need just two initial states

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CS402:Grand Quiz

Quiz Start Time: 01:44 PM, 23 June 2021

Question # 12 of 30 (Start time: 01:50:35 PM, 23 June 2021) **Total Marks:** 1

If L is a regular language, then (L')' U L will be:

Select the correct option

- none of the mentioned
- Λ
- L

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- L'

Saving

Question # 13 of 30 (Start time: 01:50:59 PM, 23 June 2021) **Total Marks:** 1

If r1 is a regular expression then (r1)* is _____.

Select the correct option

- a finite automaton
- a non-deterministic finite automaton
- a generalized transition graph
- also a regular expression

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Quiz Start Time: 01:44 PM, 23 June 2021

Question # 14 of 30 (Start time: 01:51:27 PM, 23 June 2021)

Total Marks: 1

Let FA3 be an FA corresponding to FA1FA2, then the initial state of FA3 must correspond to the initial state of

Select the correct option

- FA1 and FA2
- FA1 only

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- FA2 only
- FA1 or FA2

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Quiz Start Time: 01:44 PM, 23 June 2021

Question # 15 of 30 (Start time: 01:52:03 PM, 23 June 2021)

Total Marks: 1

Kleene's Theorem Part III expresses the relationship between _____.

Select the correct option

- TG and RE
- RE and FA

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- FA and TG
- FA and RE

Saving...

CS402:Grand Quiz

Quiz Start Time: 01:44 PM, 23 June 2021

Question # 16 of 30 (Start time: 01:52:41 PM, 23 June 2021)

Total Marks: 1

Suppose we have an alphabet set having four symbols, how many transitions will be there on each state of a finite automaton for any language defined over the given alphabet?

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Select the correct option

- 2
- 3
- 1
- 4

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Question # 17 of 30 (Start time: 01:53:03 PM, 23 June 2021) **Total Marks:** 1

If r1 = (aa + bb) and r2 = (a + b) then the language (aa + bb)(a + b) will be generated by _____.

Select the correct option

- (r2)(r1)
- (r1 + r2)
- (r1)(r2)

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- (r1)*

Saving...

Question # 18 of 30 (Start time: 01:53:30 PM, 23 June 2021) **Total Marks:** 1

Length of EVEN-EVEN language is _____.

Select the correct option

- Sometimes even & sometimes odd
- Even

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- Odd
- Such language doesn't exist

Saving...

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Quiz Start Time: 01:44 PM, 23 June 2021

Question # 19 of 30 (Start time: 01:54:00 PM, 23 June 2021)

Total Marks: 1

Which one of the following machine is represented as a pictorial representation with states and directed edges labeled by an input letter along with an output character?

Select the correct option

- Finite state machine
- Mealy machine
- Deterministic finite state machine
- Moore machine

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Question # 20 of 30 (Start time: 01:54:52 PM, 23 June 2021) Total Marks: 1

In Moore machine the output depends on

Select the correct option

- Nothing
- Type of input
- Present state and present input
- Only present state

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Quiz Start Time: 01:44 PM, 23 June 2021

Question # 21 of 30 (Start time: 01:55:13 PM, 23 June 2021)

Total Marks: 1

Which of the following doesn't accept any language?

Select the correct option

- Regular expression
- Finite state machine
- Moore machine

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- Deterministic finite state machine

Saving

Question # 22 of 30 (Start time: 01:55:37 PM, 23 June 2021) **Total Marks:** 1

Given the language L = {ab, aa, baa}, which of the following strings are in L*?
1) abaabaaabaa
2) aaaabaaaa
3) baaaaabaaaab
4) baaaaabaa

Select the correct option

- 1, 2 and 4

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- 1, 3 and 4
- 2, 3 and 4
- 1, 2 and 3

Question # 23 of 30 (Start time: 01:56:10 PM, 23 June 2021)

Total Marks: 1

Closure of an FA is the same as _____ of an FA with itself except that the initial state of the required FA is a final state as well.

Select the correct option

- Intersection
- Concatenation
- Union
- Sum

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CS402:Grand Quiz

Quiz Start Time: 01:44 PM, 23 June 2021

Question # 24 of 30 (Start time: 01:56:41 PM, 23 June 2021)

Total Marks: 1

The language of all strings defined over alphabet set = {x, y} having triple x's or triple y's will have the minimum strings with length of:

Select the correct option

- 2
- 3

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- 4
- 1

MC210201431: MUHAMMAD KHALEEQ

Time Left 71 sec(s)

CS402:Grand Quiz

Quiz Start Time: 01:44 PM, 23 June 2021

Question # 25 of 30 (Start time: 01:57:12 PM, 23 June 2021) Total Marks: 1

The language having even number of a's and even number of b's defined over S = {a, b} is called _____.

Select the correct option

- PALINDROME
- FACTORIAL
- EVEN-EVEN

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- ODD-ODD

Question # 26 of 30 (Start time: 01:57:41 PM, 23 June 2021) **Total Marks:** 1

For a given Moore Machine, the input string is '101010', thus the output string would be of length:

Select the correct option

- Length of input string - 1
- Length of input string - 2
- Length of input string + 2
- Length of input string + 1

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Question # 27 of 30 (Start time: 01:58:07 PM, 23 June 2021) **Total Marks:** 1

Let L be the language of all strings, defined over Σ = {0,1}, ending in 10. Which of the following strings are indistinguishable with respect to L with z being 0?

Select the correct option

- 110, 101
- 111, 101
- 100, 101
- 010, 101

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CS402:Grand Quiz

Quiz Start Time: 01:44 PM, 23 June 2021

Question # 28 of 30 (Start time: 01:58:36 PM, 23 June 2021)

Total Marks: 1

Strings x,y,z belongs to Σ^* such that $xz \in L$ but $yz \notin L$ where $L \subseteq \Sigma^*$ are:

Select the correct option

- (●) Distinguishable

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- ( ) Undetermined
- ( ) Indistinguishable
- ( ) Both distinguishable and indistinguishable

CS402:Grand Quiz

Quiz Start Time: 01:44 PM, 23 June 2021

Question # 29 of 30 (Start time: 01:59:18 PM, 23 June 2021)

Total Marks: 1

Let FA1 accepts many strings and FA2 accepts no string, then FA1+FA2 will be equal to:

Select the correct option

- FA2-FA1
- FA2
- FA1

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- (FA2)*

CS402:Grand Quiz

Quiz Start Time: 01:44 PM, 23 June 2021

Question # 19 of 30 (Start time: 01:54:00 PM, 23 June 2021)

Total Marks: 1

Which one of the following machine is represented as a pictorial representation with states and directed edges labeled by an input letter along with an output character?

Select the correct option

- Finite state machine
- Mealy machine
- Deterministic finite state machine
- Moore machine

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